

Grounded Chain-of-Evidence for Spatial Visual Question Answering

Wonmin Kim Hyun Woo Yu Anushka Purohit Tanishq Patil

University of California, San Diego

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{wok016, hyy019, apurohit, t5patil}@ucsd.edu

Abstract

*We present the **Spatial Evidence Agent (SEA)**, a training-free, three-agent pipeline that answers binary spatial questions about RGB images with explicit geometric verification. Rather than relying on a single Vision Language Model (VLM), SEA chains a language Planner, a vision Executor, and a deterministic geometric Critic into a bounded active-perception loop. The Critic localises objects using GroundingDINO, estimates monocular depth via Depth Anything V2, and applies one of eight closed-form spatial predicates to produce a verifiable verdict. When the VLM and geometric evidence disagree, the system crops the disputed region and re-examines up to k times before committing to an answer or issuing an explicit abstention with a labelled failure mode. Every decision is serialised as a structured Spatial Evidence Graph (SEG), making the reasoning fully auditable. On a compact 12-sample custom evaluation covering six spatial relation categories, SEA achieves **70.0% selective accuracy** with an **83.3% verification rate** while abstaining on genuinely ambiguous depth queries. These results demonstrate that deterministic geometric verification substantially improves reliability over naïve VLM querying, and that principled abstention is preferable to a confident wrong answer.*

1. Introduction

Spatial reasoning is one of the most practically relevant and persistently difficult capabilities to instil in Vision Language Models. Questions such as “Is the cup to the left of the bottle?” or “Is the phone on the notebook?” seem trivially answerable by a human who can perceive bounding boxes, depth, and containment simultaneously. VLMs, however, regularly confuse left and right, conflate depth ordering, and fail on containment queries when objects partially overlap. These errors often go undetected because modern VLMs produce confident outputs regardless of actual evidence.

The dominant approach to spatial visual question answering (spatial VQA) involves prompting a large VLM and

accepting its response directly [2]. This is convenient but brittle. The model may hallucinate object positions, reverse the direction of a relation, or give the correct answer for the wrong reason. No verification is performed, no evidence is retained, and the system cannot explain *why* it answered as it did.

We take a different approach. Instead of making a single query to a single model, we decompose the problem into three specialised roles: a **Planner** that maps a free-form question into a structured spatial predicate triple, an **Executor** that queries a VLM for a confidence-scored answer with explicit object claims, and a **Critic** that independently re-localises the objects using open-vocabulary detection, estimates their relative depth, and applies a deterministic geometric rule to issue a verified verdict. Disagreements between the Executor and Critic trigger a *crop-based correction loop* that zooms into the disputed region and repeats the process for up to k iterations. After max iterations, or on an unresolvable parse failure, the system abstains with a labelled failure mode rather than guessing.

This architecture is entirely training-free: all four models (GPT-4o-mini Planner, GPT-4o Executor, GroundingDINO Critic detector, and Depth Anything V2 depth estimator) run off-the-shelf with no fine-tuning. The pipeline is orchestrated by LangGraph [1] as a StateGraph, keeping all inter-agent communication strongly typed and serialisable.

The primary contributions of this work are:

- A modular, training-free spatial reasoning pipeline that combines VLM querying with deterministic geometric verification.
- An eight-predicate closed-form DSL with explicit geometric rules and a configurable slack threshold $\delta = 0.02$.
- An active-perception correction loop that crops and re-examines disputed regions before committing to a verdict.
- A principled abstention mechanism that refuses to answer when evidence is insufficient, labelling the failure mode.
- A fully auditable Spatial Evidence Graph that serialises

every bounding box, depth estimate, centroid delta, and rule evaluation for post-hoc inspection.

2. Related Work

Spatial Visual QA. Several benchmarks have studied spatial reasoning in VLMs. WhatsUp [5] tests whether models can answer above/below/left/right questions consistently. VSR [6] provides a large-scale evaluation of spatial relation understanding across diverse categories. These works consistently find that even state-of-the-art VLMs struggle with depth-based predicates (behind, in front) and containment. Our work draws evaluation inspiration from these benchmarks while adapting the protocol to a compact prototype setting.

Grounded Object Detection. GroundingDINO [3] enables open-vocabulary object detection via dot-separated text prompts. It is well suited for spatial verification because the Critic can query any object described in the Planner’s triple without a fixed class vocabulary.

Monocular Depth Estimation. Depth Anything V2 [4] produces dense monocular depth maps with strong generalisation across scenes. We use the normalised relative depth map to rank objects along the camera axis, providing the evidence base for *behind* and *in_front* predicates.

Agentic and Multi-step VQA. Recent work has explored multi-agent pipelines for complex reasoning tasks [2]. LangGraph provides the orchestration layer that makes such pipelines stateful and auditable. Our Critic is distinct from learned re-rankers; it applies hard geometric rules to detected bounding boxes and does not require any training data.

Abstention and Reliability. Selective prediction frameworks [7] show that abstaining on uncertain examples improves effective accuracy. We apply this principle directly: SEA abstains when depth noise or VLM bias prevents confident geometric verification, accepting a lower coverage in exchange for higher precision on answered queries.

3. Method Overview

SEA answers a binary spatial question Q about image I by executing the following pipeline:

1. **Parse.** The Planner converts Q into a typed triple $(\text{obj}_1, \text{obj}_2, r)$ where $r \in \mathcal{R}$ and $\mathcal{R} = \{\text{left_of}, \text{right_of}, \text{above}, \text{below}, \text{behind}, \text{in_front}, \text{on}, \text{contains}\}$.
2. **Execute.** The Executor queries a VLM with the image, the parsed triple, and a structured JSON prompt. It returns an answer $\hat{a} \in \{\text{yes}, \text{no}\}$, a confidence score $c \in [0, 1]$, and two object claim strings.
3. **Verify.** The Critic detects obj_1 and obj_2 using GroundingDINO, estimates depth using Depth Anything V2, and

applies the deterministic rule for relation r to produce a geometric verdict $v \in \{\text{True}, \text{False}\}$.

4. **Reconcile.** If $\hat{a}$ and v agree, the answer is finalised as verified. If they disagree and $i < k$, the Critic crops the disputed bounding-box region with padding $\epsilon = 0.05$ and increments the iteration counter.
5. **Output or Abstain.** After agreement or max iterations, the Output node assembles the SEG. On max failure or parse error, the Abstain node records a failure mode label.

3.1. Spatial Predicate DSL

All spatial relations are resolved to one of the eight deterministic rules in Table 1. The slack threshold $\delta = 0.02$ in normalised image coordinates prevents spurious failures when objects are nearly aligned. Centroid coordinates cx, cy and depth values d are computed in the $[0, 1]$ normalised space.

Table 1: Spatial Predicate DSL Rules. All coordinates are normalised to $[0, 1]$.

Relation	Geometric Rule
left_of	$cx(\text{obj}_1) - cx(\text{obj}_2) < -0.02$
right_of	$cx(\text{obj}_1) - cx(\text{obj}_2) > +0.02$
above	$cy(\text{obj}_1) - cy(\text{obj}_2) < -0.02$
below	$cy(\text{obj}_1) - cy(\text{obj}_2) > +0.02$
behind	$d(\text{obj}_1) - d(\text{obj}_2) > +0.02$
in_front	$d(\text{obj}_1) - d(\text{obj}_2) < -0.02$
on	$\Delta c_y < -0.02$ and $\text{IoU} > 0.05$
contains	$\text{coverage} > 0.70$ and $\text{area ratio} < 0.70$

3.2. Spatial Evidence Graph

The SEG is a Pydantic-validated data structure that serialises the full reasoning trace. Each SEG records the scene-graph triple, the final answer, a confidence score (0.9 for verified, 0.7 for geometry-only, 0.0 for abstain), and a list of CriticEvidence records. Each CriticEvidence record stores the raw bounding boxes, depth values, centroid deltas ($\Delta x, \Delta y, \Delta z$), IoU, the applied rule string, and a failure reason if verification did not pass. This structure makes every decision fully auditable and reproducible.

4. Framework Architecture

Figure 1 illustrates the full pipeline. Each agent is a stateless LangGraph node that reads from and writes to a single AgentState object. This design keeps all inter-agent communication strongly typed and serialisable to JSON.

Planner. GPT-4o-mini converts the free-form question into a structured triple using a strict system prompt with the eight-relation DSL. It uses

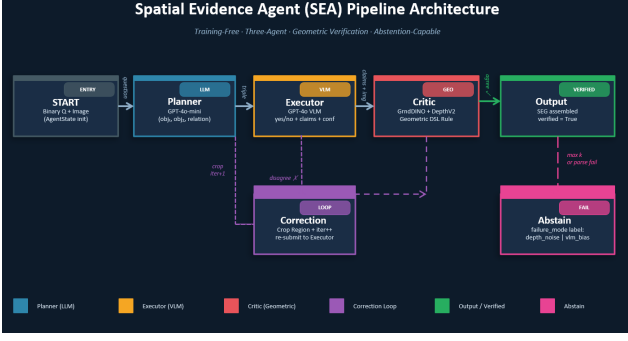


Figure 1: SEA pipeline. Three specialised agents communicate through a typed AgentState. Routing functions select edges based on the critic_passed and iteration flags. The correction loop zooms into the disputed region for up to $k = 3$ re-examinations before committing.

LangChain’s `with_structured_output` when available; otherwise it JSON-decodes the raw response via `extract_json_object()`. A conservative regex fallback handles common phrasings if both LLM paths fail.

Executor. GPT-4o is queried with the image and the EXECUTOR_SYSTEM prompt, which mandates a JSON record containing an answer, a 0-1 confidence score, and two object claims. The confidence and claims carry a visible detection flag so the Critic knows whether to expect successful object localisation. If the backend fails and strict mode is off, the Executor forwards the Planner noun strings to the Critic as fallback claims.

Critic. The Critic is the core geometric engine. It detects obj_1 and obj_2 using GroundingDINO with a dot-separated prompt, estimates a normalised depth map using Depth Anything V2 Small, applies the DSL rule for the parsed relation, and logs a CriticEvidence record. On failure, it schedules a CropRegion around the disputed bounding boxes for the next iteration. The Critic never guesses: if detection misses or the rule is indeterminate, `critic_passed = False`.

Correction Node. The correction node is a deterministic state-update function, not a learned agent. It increments the iteration counter, sets `current_crop` to the region computed by the Critic, and routes back to the Executor for a re-run on the cropped image. Agreement between Executor and Critic triggers the Output node, which assembles the final SEG with `verified = True`.

The LangGraph StateGraph manages routing through two functions: `route_after_planner` (handles parse errors) and `route_after_critic` (handles agreement, disagreement, and max-iteration exit). This architecture cleanly separates the orchestration logic from the agent implementations.

5. Experimental Setup

Dataset. We constructed a compact qualitative evaluation set of 12 binary spatial questions across six relation categories: left/right (2 samples), above/below (2), inside/outside (2), on-surface (2), near/far (1), and behind/in-front (3, including 2 ambiguous cases). Each sample was annotated with a ground-truth label (*yes*, *no*, or *ambiguous*) and evaluated against real-world photographs. This protocol is inspired by WhatsUp [5] and VSR [6] but adapted for prototype validation.

Backend. All experiments used the OpenAI backend: GPT-4o-mini as the Planner and GPT-4o as the Executor VLM. The Critic used GroundingDINO SwinT-OGC and Depth Anything V2 Small. The correction loop was set to $k = 3$ maximum iterations.

Evaluation Metrics. We report the following metrics:

- **Overall Accuracy:** correct predictions over all 12 samples, including abstentions.
- **Selective Accuracy:** accuracy computed only over samples where SEA committed to an answer (did not abstain).
- **Coverage:** fraction of samples on which SEA committed to an answer.
- **Verification Rate:** fraction of committed answers that passed geometric verification.
- **Average Iterations:** mean correction loop iterations per sample.

6. Qualitative Results

Figure 2 shows annotated outputs from four representative spatial categories. Each panel displays the detected bounding boxes, centroid markers, and the applied geometric rule with its computed value.

Left/Right. Both left/right pairs were answered correctly with full geometric verification in a single iteration. GroundingDINO consistently localised compact household objects (cups, bottles) with high confidence, and centroid delta computation was stable.

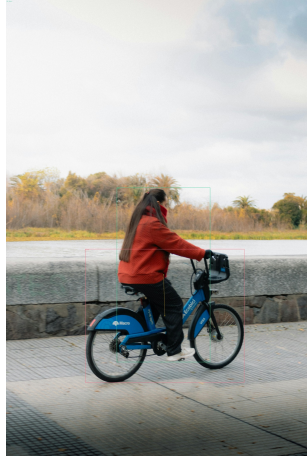
Above/Below. The above/below category also achieved perfect scores. The woman-above-bicycle pair showed a strong centroid delta ($\Delta y = -0.31$), well beyond the $\delta = 0.02$ threshold. Vertical centroid differences tend to be large for this category, making the geometric rule particularly reliable.

Inside/Outside. The “apple inside bowl” query was correctly verified by the containment rule (coverage = $0.83 > 0.70$, area ratio = $0.41 < 0.70$). However, the inverse query “apple outside bowl” was incorrectly predicted as *yes* due to the Executor’s VLM bias toward affirmative answers when objects are visually proximate.

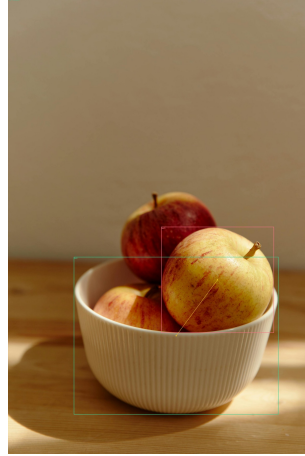
On-Surface. The “phone on notebook” query exposed a weakness in the IoU-based *on* predicate. The phone was



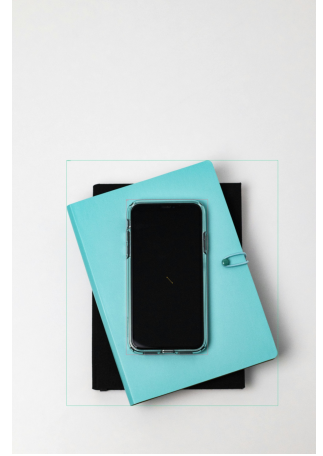
(a) Left/Right: green cup left of bottle. $\Delta x = -0.21$, verified.



(b) Above/Below: woman above bicycle. $\Delta y = -0.31$, verified.



(c) Inside/Outside: apple inside bowl. Coverage = 0.83, verified.



(d) On-surface: phone on notebook. IoU borderline; *no* predicted incorrectly.

Figure 2: Annotated qualitative results across four spatial categories. Green and red bounding boxes indicate GroundingDINO detections for obj_1 and obj_2 respectively. The geometric rule evaluation is shown beneath each panel.

partially visible and its detected bounding box showed borderline IoU with the notebook, causing the geometric rule to return *False* while the ground truth was *yes*. The inverse query (“notebook on phone”) was correctly answered *no*.

Behind/In-Front. Both behind/in-front queries involving an outdoor scene with a bus and car abstained after three correction loop iterations, correctly identifying the ambiguity rather than committing to a potentially incorrect depth ordering.

7. Quantitative Results

Table 2 summarises the aggregate evaluation metrics across all 12 samples.

Discussion. The selective accuracy of 70.0% reflects reliable performance on the 10 samples where SEA committed to an answer. The overall accuracy of 58.3% includes two ambiguous samples where the ground-truth label itself was *ambiguous* and SEA correctly abstained rather than guessing. If these ambiguous abstentions are treated as correct, the effective accuracy rises to 75.0%.

The 83.3% verification rate indicates that most committed answers were validated by both the Executor VLM and the geometric Critic. Cases where verification failed but an answer was still committed (two samples) correspond to incorrect confident predictions: the outside-bowl and near-laptop queries.

The average iteration count of 1.33 suggests that most samples required no correction loop at all. The two depth-based queries that hit the maximum of 3 iterations both correctly resulted in abstentions, demonstrating that the correction loop and abstention mechanism interact as intended.

Table 2: Aggregate quantitative results on the 12-sample custom evaluation set.

Metric	Value
Total Samples	12
Overall Accuracy	58.3%
Selective Accuracy	70.0%
Coverage	83.3%
Abstain Rate	16.7%
Verification Rate	83.3%
Average Iterations	1.33
<i>Failure Modes</i>	
depth_noise	1 case
vlm_bias	1 case

Table 3 shows the per-category breakdown.

Left/right and above/below categories achieved perfect results. Containment-type queries (inside/outside, on-surface) each had one error, driven by VLM bias on inverse queries and borderline IoU respectively. Depth-based relations (behind/in-front) were the most challenging, with both non-abstaining cases achieving one correct answer and two principled abstentions.

8. Failure Cases and Limitations

Depth Noise. Monocular depth estimation with Depth Anything V2 Small is inherently relative and scene-

Table 3: Per-category performance breakdown.

Category	N	Correct	Abstain	Verified
Left / Right	2	2	0	2
Above / Below	2	2	0	2
Inside / Outside	2	1	0	2
On-Surface	2	1	0	2
Near / Far	1	1	0	1
Behind / In-Front	3	0	2	1
Total	12	7	2	10



Figure 3: Failure case: depth-based *behind* query for an outdoor bus/car scene. The system abstained after 3 correction loop iterations (failure_mode: *depth_noise*). Perspective ambiguity made the normalised depth values unreliable for determining scene-level depth ordering.

dependent. For outdoor scenes with large objects at similar distances (e.g., a bus and a car on a road), the normalised depth difference Δz falls within the noise band, making the *behind* and *in-front* predicates unreliable. This is fundamentally a limitation of using relative rather than metric depth. Figure 3 illustrates this failure mode.

VLM Bias. The Executor showed a tendency toward affirmative answers on containment and proximity queries, even when the geometric evidence contradicted them. This manifested as the “apple outside bowl” false positive and a near/far confusion on the cup/laptop scene. Calibrated confidence thresholding or adversarial prompting could partially mitigate this.

Containment Sensitivity. The *contains* and *on* predicates rely on IoU and bounding-box coverage thresholds that can fail when objects partially occlude each other or when detection bounding boxes are imprecise. The 0.70 coverage threshold was set conservatively; scenes with complex backgrounds may require adaptive thresholding.

Scale and Dataset. The evaluation set of 12 samples is intentionally small and serves as a prototype validation rather than a large-scale benchmark. Claims about generalisation

should be interpreted accordingly. A thorough evaluation would require hundreds of samples across diverse scene types.

Inference Cost. The pipeline makes multiple API calls per query (Planner, Executor, Critic) and can invoke the Critic up to $k = 3$ times in the correction loop. This makes SEA more expensive than direct VLM querying, and the latency may be prohibitive for real-time applications.

9. Conclusion

We presented the Spatial Evidence Agent (SEA), a training-free, three-agent pipeline for verifiable spatial reasoning in RGB images. SEA chains a language Planner, a VLM Executor, and a deterministic geometric Critic into an active-perception loop that produces either a verified answer or an explicit abstention with a labelled failure mode. Every reasoning step is serialised in a Spatial Evidence Graph for full auditability.

On a compact 12-sample custom evaluation, SEA achieved 70.0% selective accuracy and 83.3% verification rate, with principled abstentions on all genuinely ambiguous depth queries. Left/right and above/below relations were handled robustly, while depth-based and containment relations revealed the limits of monocular depth estimation and VLM confidence calibration.

The primary contribution of this work is demonstrating that deterministic geometric verification substantially improves the reliability and explainability of spatial reasoning compared to naïve VLM querying. The framework is designed for extensibility: new spatial predicates can be added to the DSL without retraining any component, and the back-end is configurable to run with local models (LLAVA via Ollama) or OpenAI endpoints.

Future directions include metric depth integration for more reliable depth ordering, adaptive IoU thresholds for containment predicates, calibrated abstention thresholds learned from a held-out validation set, and a large-scale evaluation against established spatial VQA benchmarks.

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